PROBLEM 15 | Problems T2 defects.pdf | Crystal Defects & Solid Solutions

Vacancy Fraction in Aluminium near the Melting Point

Original statement · Problems T2 defects.pdf

1. The fraction of vacancies in an aluminum lattice is 2.29×10^{-5} *at* $400^{\circ}\mathrm{C}$*. Calculate the fraction of vacancies at* $660^{\circ}\mathrm{C}$ *(just before the melting point).* $R = 8.31\ \mathrm{J/(mol \cdot K)}$*. (Solution:* 4.53×10^{-4}*)*

Solution: 4.53×10^{-4}

PHASE 1: HYPOTHESIS & PARAMETERS

Input Data:

- Material: Pure aluminium (Al, FCC lattice).
- Temperature 1: $T_1 = 400^{\circ}\mathrm{C} = 400 + 273.15 = 673.15\ \mathrm{K}$.
- Vacancy fraction at T_1:

$$\left(\frac{n_v}{N}\right)_1 = 2.29 \times 10^{-5}$$

- Temperature 2: $T_2 = 660^{\circ}\mathrm{C} = 660 + 273.15 = 933.15\ \mathrm{K}$ (temperature just below the melting of aluminium, $T_m \approx 660.3^{\circ}\mathrm{C}$).
- Universal gas constant: $R = 8.31\ \mathrm{J/(mol \cdot K)}$.

PHASE 2: FUNDAMENTAL PHYSICS & FORMULA SELECTION

The thermodynamic equilibrium concentration of vacancies in a metallic lattice follows the Boltzmann distribution [Session 4 Slide 16]:

$$\frac{n_v}{N} = \exp\left(-\frac{\Delta H_v}{RT}\right)$$

where ΔH_v is the enthalpy of formation of one mole of vacancies in the lattice.

Solution Strategy:

1. With the data at T_1, determine the vacancy formation enthalpy ΔH_v:

$$\ln\left(\frac{n_v}{N}\right)_1 = -\frac{\Delta H_v}{R \cdot T_1} \implies \Delta H_v = -R \cdot T_1 \cdot \ln\left(\frac{n_v}{N}\right)_1$$

2. With ΔH_v constant (independent of temperature in this range), compute the new fraction at T_2:

$$\left(\frac{n_v}{N}\right)_2 = \exp\left(-\frac{\Delta H_v}{R \cdot T_2}\right)$$

Or, combining into a single two-point relation:

$$\ln\left(\frac{(n_v/N)_2}{(n_v/N)_1}\right) = -\frac{\Delta H_v}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$$

`PHASE 3: ANALYTICAL DERIVATION & CALCULATION STEP-BY-STEP`

1. Formation Enthalpy (ΔH_v):

- Natural logarithm of the initial fraction:

$$\ln(2.29 \times 10^{-5}) = -10.68437$$

- Solving for ΔH_v:

$$\Delta H_v = -(8.31\ \text{J/mol} \cdot \text{K}) \times (673.15\ \text{K}) \times (-10.68437)$$

$$\Delta H_v = 5593.8765 \times 10.68437 = 59767.1\ \text{J/mol} \approx \mathbf{59.77}\ \text{kJ/mol}$$

(In atomic units: $E_v = \frac{59767.1}{6.022\times10^{23}\times1.602\times10^{-19}} \approx 0.62$ eV/atom, consistent with the literature for aluminium).

---

2. Vacancy Fraction at 660°C ($T_2 = 933.15$ K):

- Boltzmann exponent at T_2:

$$\frac{\Delta H_v}{R \cdot T_2} = \frac{59767.1 \text{ J/mol}}{(8.31 \text{ J/mol} \cdot \text{K}) \times (933.15 \text{ K})} = \frac{59767.1}{7754.4765} = 7.70743$$

- Equilibrium vacancy fraction:

$$\left(\frac{n_v}{N}\right)_2 = \exp(-7.70743) = \mathbf{4.49 \times 10^{-4}}$$

Check with $T = 273 + t$ (i.e. $T_1 = 673$ K, $T_2 = 933$ K): $\Delta H_v = 8.31 \times 673 \times 10.68437 = 59754$ J/mol and $(n_v/N)_2 = \exp(-59754/(8.31 \times 933)) = \exp(-7.7069) = 4.50 \times 10^{-4}$.

Discrepancy with the official solution. The official key gives 4.53×10^{-4}. Recomputing from the given data (2.29×10^{-5} at 400°C, $R = 8.31$ J/(mol K)) gives $\exp(-7.7074) = 4.49 \times 10^{-4}$ with $T = 273.15 + t$, and $\exp(-7.7069) = 4.50 \times 10^{-4}$ with $T = 273 + t$ ($\Delta H_v = 59.75$ kJ/mol). The key's 4.53×10^{-4} ($\approx 0.8\%$ higher) is not reproduced by either convention; no value was adjusted.

PHASE 4: PHYSICAL INTERPRETATION & VERIFICATION

- **Exponential Thermal Increase:** Going from 400°C to 660°C ($\Delta T = 260^\circ$C), the vacancy concentration is multiplied by a factor of:

$$\frac{4.49 \times 10^{-4}}{2.29 \times 10^{-5}} \approx \mathbf{19.6} \text{ times}$$

- **Physical Limit of Real Crystals:** The fraction reached near melting ($4.49 \times 10^{-4} \approx$ 1 vacancy per 2200 atoms) satisfies the general rule quoted in the official slide [Session 4 Slide 16]:

$$\frac{n_v}{N} \sim 10^{-4} \text{ maximum}$$

This high density of thermal vacancies near the melting point is responsible for the drastic increase in the rate of atomic diffusion in homogenisation and aeronautical sintering processes.

Study notes written from the official course material (Problems T2_CrystStruct.pdf; Problems T2 defects.pdf).
Online version: subjects/aerospace-materials-1/problemas/topic-2-problems.html